Cool macrofauna title

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Abstract
This document is only a demo explaining how to use the Stylish Article template.

Keywords
soil macrofauna, Paracou, Amazon forest, ecology

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1. Introduction

This template allows writing stylish articles in Markdown with Quarto[1]. It directly produces well-formatted PDF articles for self-archiving or in other formats, for example HTML.

2. Markdown

Markdown is a very simple language for producing various types of documents: HTML, PDF, and Word among others. Its documentation is available on the Quarto website[2].

This document was made RStudio, but other tools are available[3]: Quarto processes the Markdown code, passes it to Pandoc for transformation into LaTeX based on the Stylish Article template provided in this Quarto extension; finally LaTeX compiles it into PDF.

2.1 Motivation

Markdown is very easy to learn.

Markdown allows integrating your R, Python and more code for a *reproducible* result.

Markdown allows to produce, without rewriting the text, a document in different formats: HTML, LaTeX or Word for example.

3. Code

The main features of Markdown are summarized here. The full documentation is on-line[4].

3.1 R and other code

R code is included in code chunks:

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

Similar chunks support other languages such as Python. This template focuses on R but can be used with any other code language supported by Quarto.

[1] https://quarto.org/
[2] https://quarto.org/docs/authoring/markdown-basics.html
[3] https://quarto.org/docs/get-started/

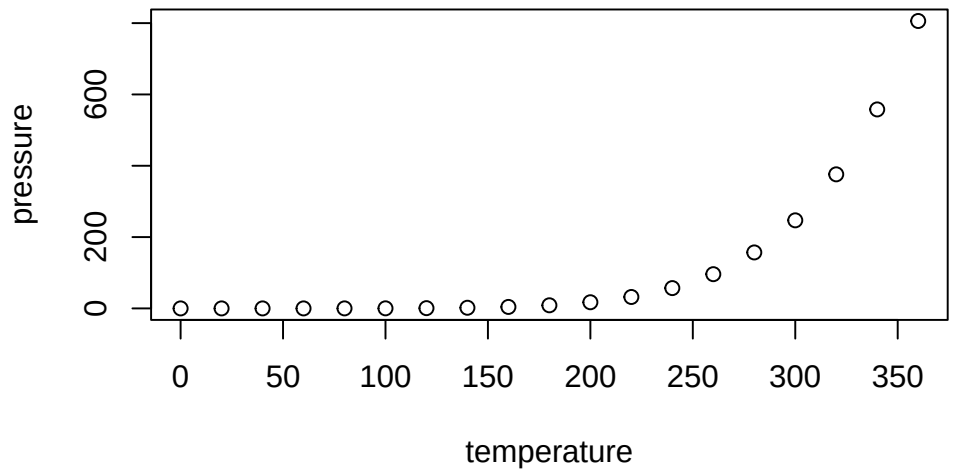


Figure 1. Comparison of motorway and city driving ranges for 38 car models.

3.2 Figures

Figures can be created by R code (Figure 1) in a division whose name starts with `#fig-`. Cross-references are made with the command `@label`, e.g.: `@fig-pressure`.

In PDF, a figure can use the full width of the page by adding the options `fig-env="figure*"` to the header of the division.

Images stored in a file are integrated into a piece of code by the `include_graphics()` function, see Figure 2.



Figure 2. This is a logo.

Systematically place these files in the `images` folder for the automation of GitHub pages.

3.3 Tables

The horizontal `-` and vertical separators `|` allow you to draw a table according to Markdown syntax, but this is not the best method in R.

[4] https://quarto.org/docs/authoring/markdown-basics.html/

Table 1. Table created by R in a data frame.

Sepal length	Sepal width	Petal length	Petal width	Species
5.1	3.5	1.4	0.2	setosa
4.9	3.0	1.4	0.2	setosa
4.7	3.2	1.3	0.2	setosa
4.6	3.1	1.5	0.2	setosa
5.0	3.6	1.4	0.2	setosa
5.4	3.9	1.7	0.4	setosa

Tables can also be produced by R code. The content of the table is in a dataframe. The `gt()` function from package *gt* formats it. Refer to the documentation of the package for details about formatting the output.

By default, the tables is contained in its column.

A full-width table is Table 2. It is obtained with the option `fig-env="table*"` in the header of the division.

Note that tables can't be shown on the first page of the PDF output of the Stylish Article template: they would conflict with the table of contents.

3.4 Lists

Lists are indicated by `*`, `+` and `-` (three hierarchical levels) or numbers `1.`, `i.` and `A.` (numbered lists). Indentation of lists indicates their level: `*`, `+` and `-` may be replaced by `-` at all levels, but four spaces are needed to nest a list into another.

- First element of a list
  - sub-list

Table 2. Large table with maths, on two columns in PDF.

Treatment	Timber	Thinning	Fuelwood	%AGB lost
Control				0
T1	DBH $\geq$ 50 cm, commercial species, $\approx$ 10 trees/ha			[12% – 33%]
T2	DBH $\geq$ 50 cm, commercial species, $\approx$ 10 trees/ha	DBH $\geq$ 40 cm, non-valuable species, $\approx$ 30 trees/ha		[33% – 56%]
T3	DBH $\geq$ 50 cm, commercial species, $\approx$ 10 trees/ha	DBH $\geq$ 50 cm, non-valuable species, $\approx$ 15 trees/ha	40 cm $\leq$ DBH $\leq$ 50 cm, non-valuable species, $\approx$ 15 trees/ha	[35% – 56%]

- Second element
- Continuation of the list

Leave an empty line before and after the list, but not between its items.

3.5 Math

Equations in LaTeX format can be inserted inline, like $A = \pi r^2$ or isolated like

$$e^{i\pi} = -1.$$

They can be numbered, see Equation 1, after adding them a label:

$$A = \pi r^2. \tag{1}$$

3.6 Cross-references

Figures and tables have a label declared in their division `tbl-xxx` or `fig-yyy`, starting with `fig-` or `tbl-`.

For equations, the label is added manually by the code `{#eq-xxx}` after the end of the equation.

Sections can be tagged by ending their title with `{#sec-yyy}`.

In all cases, the call to the reference is made by `@`.

3.7 Bibliography

Bibliographic references included in the `.bib` file declared in the document header can be called by `[@CitationKey]`, in parentheses (Xie et al. 2018), or without square brackets, in the text, as Xie (2016).

The bibliographic style can be specified, by adding the line

```
csl: file_name.csl
```

in the document header and copying the *.csl* style file to the project folder. More than a thousand styles are available[5] and their URL can be used instead of copying the file, e.g.:

```
csl: https://www.zotero.org/styles/xxx
```

3.8 Forcing line breaks

Hyphenation is handled automatically in LaTeX. If a word is not hyphenated correctly, add its hyphenation in the preamble of the file with the command `hyphenation` (words are separated by spaces, hyphenation locations are represented by dashes).

If LaTeX can't find a solution for the line break, for example because some code is too long a non-breaking block, add the LaTeX command `\break` to the line break location. Do not leave a space before the command. The HTML document ignores LaTeX commands.

3.9 Languages

Languages are declared in the document header.

The main language of the document (`lang`) changes the name of some elements, such as the table of contents. The change of language in the document (one of `otherlangs`) is managed in LaTeX (but not fully in HTML).

For a single word, to ensure correct hyphenation in LaTeX, use the following command, ignored un HTML:

```
\foreignlanguage{italian}{ciao}
```

For a paragraph, to also ensure correct quotes and punctuation spacing, use

```
::: {lang=fr}
"Bonjour": en français!
:::
```

to obtain:

« Bonjour » : en français !

[5] https://github.com/citation-style-language/styles

The current language has an effect only in LaTeX output: a space is added before double punctuation in French, the size of spaces is larger at the beginning of sentences in English, etc.

3.10 Text color

This template includes the "color-text" filter for Quarto. It makes colored text possible using the `[content]{color=<name>}` syntax. This is a red example. Defined colors are blue, green and red. To add more colors, they must be defined in the document header in `format:`, `stylisharticle-pdf:`, `header-includes:` . This is the code for "grey":

```
\definecolor{grey}{RGB}{191, 191, 191}
```

They must also be included in a `.css` file declared in `format:`, `stylisharticle-html:`

```
css: colors.css
```

In this css, colors are declared by

```
.color-grey {color: rgb(191, 191, 191);}
```

4. .gitignore

Copy this content to the `.gitignore` file:

```
# RStudio files and folders
.Rproj.user
.Rhistory
.RData
.Ruserdata

# Mac OS
.DS_Store

# Output
/*.html
/*.pdf
/*.tex
# Intermediate files
/nul
/*.aux
/*.log
/*.toc
# Working folders
/*_files/
/*_cache/

# Template (copied from _extensions)
/stylisharticle.cls
```

5. GitHub Action

Create a file named `quarto_publish.yml` in the folder `.github/workflows` (to be created) to activate continuous integration:

```
on:
  push:
    branches: [main, master]

name: Render and Publish

jobs:
  build-deploy:
    runs-on: macOS-latest
    permissions:
      contents: write
    steps:
      - name: Check out repository
        uses: actions/checkout@v7
      - name: Set up R
        uses: r-lib/actions/setup-r@v2
      - name: Set up Quarto
        uses: quarto-dev/quarto-actions/setup@v2
        env:
          GH_TOKEN: ${{ secrets.GITHUB_TOKEN }}
        with:
          tinytex: true
      - name: Install dependencies
        run: |
          options(pkgType = "binary")
          options(install.packages.check.source = "no")
          install.packages(c("rmarkdown", "knitr"))
        shell: Rscript {0}
      - name: Publish to GitHub Pages
        uses: quarto-dev/quarto-actions/publish@v2
        with:
          path: index.qmd
          target: gh-pages
        env:
          GITHUB_TOKEN: ${{ secrets.GITHUB_TOKEN }}
```

6. Caveats

Quarto does not yet support all features of R Markdown. Unsolved issues are listed here, as of version 1.9:

- The `column` options in code chunks or `{.column-xxx}` div's can't be used in this template. They interfere with the two-column LaTeX layout. The template relies on other techniques to print full-width figures and tables.

References

Xie, Yihui. 2016. *Bookdown: Authoring Books and Technical Documents with R Markdown*. Chapman; Hall/CRC. https://github.com/rstudio/bookdown.

Xie, Yihui, J. J. Allaire, and Garrett Grolemund. 2018. *R Markdown: The Definitive Guide*. Chapman; Hall/CRC. https://bookdown.org/yihui/rmarkdown.